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Adapting hotword recognition based on personalized negatives

Abstract

A method for adapting hotword recognition includes receiving audio data characterizing a hotword event detected by a first stage hotword detector in streaming audio captured by a user device. The method also includes processing, using a second stage hotword detector, the audio data to determine whether a hotword is detected by the second stage hotword detector in a first segment of the audio data. When the hotword is not detected by the second stage hotword detector, the method includes, classifying the first segment of the audio data as containing a negative hotword that caused a false detection of the hotword event in the streaming audio by the first stage hotword detector. Based on the first segment of the audio data classified as containing the negative hotword, the method includes updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data that contains the negative hotword.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This U.S. patent application is a continuation of, and claims priority under 35 U.S.C. § 120 from, U.S. patent application Ser. No. 16/953,510, filed on Nov. 20, 2020. The disclosure of this prior application is considered part of the disclosure of this application and is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) This disclosure relates to adapting hotword recognition based on personalized negatives.

BACKGROUND

(2) A speech-enabled environment (e.g., home, workplace, school, automobile, etc.) allows a user to speak a query or a command out loud to a computer-based system that fields and answers the query and/or performs a function based on the command. The speech-enabled environment can be implemented using a network of connected microphone devices distributed through various rooms or areas of the environment. These devices may use hotwords to help discern when a given utterance is directed at the system, as opposed to an utterance that is directed to another individual present in the environment. Accordingly, the devices may operate in a sleep state or a hibernation state and wake-up only when a detected utterance includes a hotword. Typically, systems used to detect hotwords in streaming audio generate a probability score indicative of a probability that a hotword is present in the streaming audio. When the probability score satisfies a predetermined threshold, the device initiates the wake-up process.

SUMMARY

(3) One aspect of the disclosure provides a method for adapting hotword recognition based on personalized negatives. The method includes receiving, at data processing hardware, audio data characterizing a hotword event detected by a first stage hotword detector in streaming audio captured by a user device. The method also includes processing, by the data processing hardware, using a second stage hotword detector, the audio data to determine whether a hotword is detected by the second stage hotword detector in a first segment of the audio data. When the hotword is not detected by the second stage hotword detector in the first segment of the audio data, the method includes classifying, by the data processing hardware, the first segment of the audio data as containing a negative hotword that caused a false detection of the hotword event in the streaming audio by the first stage hotword detector. Based on the first segment of the audio data classified as containing the negative hotword, the method includes updating, by the data processing hardware, the first stage hotword detector to prevent triggering the hotword event in subsequent audio data that contains the negative hotword.

(4) Implementations of the disclosure may include one or more of the following optional features. In some implementations, the method further includes, when the hotword is not detected by the second stage hotword detector in the first segment of the audio data suppressing, by the data processing hardware, a wake-up process on the user device for processing the hotword and one or more other terms following the hotword in the streaming audio; and determining, by the data processing hardware, whether an immediate follow-up query was provided by a user of the user device after suppressing the wake-up process on the user device. In these implementations, classifying the first segment of the audio data as containing the negative hotword is further based on determining that no follow-up query was provided by the user of the user device after suppressing the wake-up process.

(5) In some examples, when the hotword is detected by the second stage hotword detector in the first segment of the audio data, the method further includes processing, by the data processing hardware, a second segment of the audio data that follows the first segment of the audio data to determine whether the second segment of the audio data is indicative of a spoken query-type

utterance. In these examples, when the second audio segment of the audio data is not indicative of the spoken query-type utterance, the method also includes: classifying, by the data processing hardware, the first segment of the audio data as containing the negative word; and based on the first segment of the audio data classified as containing the negative hotword, updating, by the data processing hardware, the first stage hotword detector to prevent triggering the hotword event in subsequent audio data that includes the negative hotword. In some implementations, the method further includes, when the second audio segment of the audio data is not indicative of the spoken query-type utterance, determining, by the data processing hardware, whether an immediate follow-up query was provided by a user of the user device. Here, classifying the first segment of the audio data as containing the negative hotword is further based on determining that no follow-up query was provided by the user of the user device. When the second audio segment of the audio data is indicative of the spoken query-type utterance, the method may also include: receiving, at the data processing hardware, a negative interaction result indicating that a user of the user device negatively interacted with results for the spoken query type utterance provided to the user device; classifying, by the data processing hardware, based on the received negative interaction result, the first segment of the audio data as containing the negative hotword; and based on the first segment of the audio data classified as containing the negative hotword, updating, by the data processing hardware, the first stage hotword detector to prevent detecting the hotword event in subsequent audio data that contains the negative hotword.

(6) In some examples, after receiving the audio data characterizing the hotword event detected by the first stage hotword detector, the method further includes receiving, at the data processing hardware, a negative user interaction indicating user suppression of a wake-up process on the user device. Here, classifying the first segment of the audio data as containing the negative hotword is further based on the negative user interaction indicating user suppression of the wake-up process.

(7) Optionally, updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data may include providing the first segment of the audio data classified as containing the negative hotword to the user device. The user device is configured to retrain the first stage hotword detector using the first segment of audio data classified as containing the negative hotword. In some implementations, the user device is configured to retrain the first stage hotword detector by storing, in memory hardware of the user device, each instance of the first segment of the audio data classified as containing the negative hotword in memory hardware of the user device and retraining the first stage hotword detector based on an aggregation of the number of instances of the first segment of the audio data classified as containing the negative hotword stored in the memory hardware. In these implementations, the user device is further configured to, prior to retraining the first stage hotword detector, determine that a corresponding confidence score associated each instance of the first segment of the audio data classified as containing the negative hotword fails to satisfy a negative hotword threshold score and determine that the number of instances exceeds a threshold number of instances.

(8) Updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data may include providing the first segment of the audio data classified as containing the negative hotword to the user device. The user device configured to obtain an embedding representation of the first segment of the audio data and store, in memory hardware of the user device, the embedding representation of the first segment of the audio data. Additionally, the user device is configured to determine when subsequent audio data characterizing the hotword event detected by the first stage hotword detector includes the negative hotword by computing an evaluation embedding representation for the audio data; determine a similarity score between the embedding representation of the first segment of the audio data classified as the negative hotword and the evaluation embedding representation for the subsequent audio data; and when the similarity score satisfies a similarity score threshold, determine that the subsequent audio data includes the negative hotword.

(9) In some implementations, the data processing hardware resides on a server in communication with the data processing hardware and the first stage hotword detector executes on a processor of the user device. Processing the audio data to determine whether the hotword is detected by the second stage hotword detector in the first segment of the audio data may include performing automated speech recognition to determine whether the hotword is recognized in the first segment of the audio data.

(10) In some examples, the data processing resides on the user device. In these examples, the first stage hotword detector may execute on a digital signal processor (DSP) of the data processing hardware and the second stage hotword detector executes on an application processor of the data processing hardware. The first stage hotword detector may be configured to generate a probability score that indicates a presence of the hotword in audio features of the streaming audio captured by the user device and detect the hotword event in the streaming audio when the probability score satisfies a hotword detection threshold of the first stage hotword detector.

(11) Another aspect of the disclosure provides a system for adapting hotword recognition based on personalized negatives. The system includes data processing hardware and memory hardware in communication with the data processing hardware. The memory hardware stores instructions that when executed on the data processing hardware cause the data processing hardware to perform operations. The operations include receiving audio data characterizing a hotword event detected by a first stage hotword detector in streaming audio captured by a user device. The operations also include processing, using a second stage hotword detector, the audio data to determine whether a hotword is detected by the second stage hotword detector in a first segment of the audio data. When the hotword is not detected by the second stage hotword detector in the first segment of the audio data, the operations include classifying the first segment of the audio data as containing a negative hotword that caused a false detection of the hotword event in the streaming audio by the first stage hotword detector. Based on the first segment of the audio data classified as containing the negative hotword, the operations include updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data that contains the negative hotword.

(12) Implementations of the disclosure may include one or more of the following optional features. In some implementations, the operations further include, when the hotword is not detected by the second stage hotword detector in the first segment of the audio data suppressing a wake-up process on the user device for processing the hotword and one or more other terms following the hotword in the streaming audio; and determining whether an immediate follow-up query was provided by a user of the user device after suppressing the wake-up process on the user device. In these implementations, classifying the first segment of the audio data as containing the negative hotword is further based on determining that no follow-up query was provided by the user of the user device after suppressing the wake-up process.

(13) In some examples, when the hotword is detected by the second stage hotword detector in the first segment of the audio data, the operations further include processing a second segment of the audio data that follows the first segment of the audio data to determine whether the second segment of the audio data is indicative of a spoken query-type utterance. In these examples, when the second audio segment of the audio data is not indicative of the spoken query-type utterance, the operations also include classifying the first segment of the audio data as containing the negative word; and based on the first segment of the audio data classified as containing the negative hotword, updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data that includes the negative hotword. In some implementations, the operations further include, when the second audio segment of the audio data is not indicative of the spoken query-type utterance, determining whether an immediate follow-up query was provided by a user of the user device. Here, classifying the first segment of the audio data as containing the negative hotword is further based on determining that no follow-up query was provided by the user of the user device. When the second audio segment of the audio data is indicative of the spoken query-

type utterance, the operations may also include: receiving a negative interaction result indicating that a user of the user device negatively interacted with results for the spoken query type utterance provided to the user device; classifying, based on the received negative interaction result, the first segment of the audio data as containing the negative hotword; and based on the first segment of the audio data classified as containing the negative hotword, updating the first stage hotword detector to prevent detecting the hotword event in subsequent audio data that contains the negative hotword.

(14) In some examples, after receiving the audio data characterizing the hotword event detected by the first stage hotword detector, the operations further include receiving a negative user interaction indicating user suppression of a wake-up process on the user device. Here, classifying the first segment of the audio data as containing the negative hotword is further based on the negative user interaction indicating user suppression of the wake-up process.

(15) Optionally, updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data may include providing the first segment of the audio data classified as containing the negative hotword to the user device. The user device is configured to retrain the first stage hotword detector using the first segment of audio data classified as containing the negative hotword. In some implementations, the user device is configured to retrain the first stage hotword detector by storing, in memory hardware of the user device, each instance of the first segment of the audio data classified as containing the negative hotword in memory hardware of the user device and retraining the first stage hotword detector based on an aggregation of the number of instances of the first segment of the audio data classified as containing the negative hotword stored in the memory hardware. In these implementations, the user device is further configured to, prior to retraining the first stage hotword detector, determine that a corresponding confidence score associated each instance of the first segment of the audio data classified as containing the negative hotword fails to satisfy a negative hotword threshold score and determine that the number of instances exceeds a threshold number of instances.

(16) Updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data may include providing the first segment of the audio data classified as containing the negative hotword to the user device. The user device configured to obtain an embedding representation of the first segment of the audio data and store, in memory hardware of the user device, the embedding representation of the first segment of the audio data. Additionally, the user device is configured to determine when subsequent audio data characterizing the hotword event detected by the first stage hotword detector includes the negative hotword by: computing an evaluation embedding representation for the audio data; determine a similarity score between the embedding representation of the first segment of the audio data classified as the negative hotword and the evaluation embedding representation for the subsequent audio data; and when the similarity score satisfies a similarity score threshold, determine that the subsequent audio data includes the negative hotword.

(17) In some implementations, the data processing hardware resides on a server in communication with the data processing hardware and the first stage hotword detector executes on a processor of the user device. Processing the audio data to determine whether the hotword is detected by the second stage hotword detector in the first segment of the audio data may include performing automated speech recognition to determine whether the hotword is recognized in the first segment of the audio data.

(18) In some examples, the data processing resides on the user device. In these examples, the first stage hotword detector may execute on a digital signal processor (DSP) of the data processing hardware and the second stage hotword detector executes on an application processor of the data processing hardware. The first stage hotword detector may be configured to generate a probability score indicating a presence of the hotword in audio features of the streaming audio captured by the user device and detect the hotword event in the streaming audio when the probability score satisfies a hotword detection threshold of the first stage hotword detector.

(19) The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic view of an example system for classifying negative hotwords and updating a first stage hotword detector to prevent detecting hotword events in audio containing negative hotwords.
- (2) FIG. 2 is a schematic view of a hotword detection architecture.
- (3) FIG. 3 is a schematic view of an example negative hotword classifier of the system of FIG. 1.
- (4) FIG. 4 is a schematic view of a user device storing classification results for audio data classified as containing a negative hotword.
- (5) FIG. 5 is a schematic view of an example user device identifying the presence of a personal negative hotword in captured audio data to prevent triggering a hotword event in the audio data.
- (6) FIG. 6 is a flowchart of an example arrangement of operations for classifying personalized negative hotwords and updating a first stage hotword detector to prevent triggering a hotword event in audio data containing a personalized negative hotword.
- (7) FIG. 7 is a schematic view of an example computing device that may be used to implement the systems and methods described herein.
- (8) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

- (9) Speech-based interfaces such as digital assistants are becoming increasingly prevalent across a variety of devices including, without limitation, mobile phones and smart speakers/displays that include microphones for capturing speech. The general way of initiating voice interaction with a speech-enabled device is to speak a fixed phrase, e.g., a hotword, that when detected by the speech-enabled device in streaming audio, triggers the speech-enabled device to initiate a wake-up process to begin recording and processing subsequent speech to ascertain a query spoken by the user. Thus, as a primary entry point for a speech-based interface, it is critical that hotword detection/recognition works reliably both in terms of recall, and also precision, so that a number of false wake-up events is minimized.
- (10) A false negative (also referred to as ‘false rejection’) refers to not detecting a hotword spoken in streaming audio that was spoken by the user when intending to interact with the speech-based interface (e.g., digital assistant). Here, the speech-enabled device fails to react to the user and requires the user to attempt to invoke the interface again by speaking the hotword again, often louder and with different enunciation, to ensure that the hotword is detected. On the other hand, a false positive (also referred to as ‘false acceptance’) refers to detecting a hotword in streaming audio when the streaming audio did not actually contain the hotword, generally due to the streaming audio containing a word/phrase that, when spoken, sounds phonetically similar to the hotword. The false positive causes the speech-enabled device to initiate the wake-up process even though the user did not intend to invoke the system, thereby surprising and/or confusing the user by reacting when the speech-enabled device should have remained in a sleep state.
- (11) A cascade hotword detection architecture incorporates a first stage hotword detector that runs on device to detect the presence of a hotword in streaming, and a second stage hotword detector that confirms the presence of the hotword detected by the first stage hotword detector. The second stage hotword detector is associated with higher accuracy in detecting the presence of hotwords in streaming audio, and thus, includes higher power requirements than the first stage hotword detector. Often, the second stage hotword detector is implemented on a server in communication

with the first stage hotword detector implemented on the speech-enabled device. Even during a partial false positive, where the first stage hotword detector detects the presence of a hotword locally but the second stage hotword detector implemented at the server rejects the presence of the hotword, has a negative effect on user experience even though the server ultimately suppresses the wake-up process. That is, the detection of the hotword by the first stage hotword detector still causes the device to wake-up and connect to the server which is noticeable to the user (e.g., visible notification or flashing light) and is further undesirable from the privacy and power-preserving perspectives. Accordingly, eliminating the occurrence of partial false positive instances is desirable to improve user experience.

(12) Conventionally, speech-enabled devices all use a same, fixed hotword model for all users of a given language (or locale) that is updated periodically with new versions pushed from server to device. That is, the same hotword model is used to detect hotwords in streaming audio for all users despite the existence of huge variations across user speech, accents, vocabulary, and/or acoustic environments in which the speech-enabled devices are operating. As a result, it is nearly impossible to implement stringent precision/recall requirements for detecting hotwords when a single hotword model is shared across all users of a given language/locale.

(13) For a given user and/or environment, hotword false positive instances are very likely to cluster together. In a non-limiting example, a particular user speaking the term “poodle” may cause a hotword detection model on a speech-enabled device to incorrectly detect the presence of the designated hotword “Hey Google”, whereas a different user may cause the same hotword detection model implemented on another speech-enabled device to detect the designated hotword when that user speaks “doodle”. The variation in these false positive instances across different users can be attributed to pronunciation differences of the users and/or frequency of those terms in the respective vocabularies of those users. Since the same false positive instances are likely to reoccur based on similar acoustic patterns for a same user in a same environment, a hotword detector should ideally learn to adapt to avoid repeating the same false positives over and over when the user speaks a given with a pronunciation similar to the designated hotword.

(14) Implementations herein are directed toward personalizing a hotword detector on a speech-enabled device of a user based on specific terms classified as negative hotwords which caused previous instances of hotword detection false positives. A specific term classified as a negative hotword may be user-specific such that any hotword detection is suppressed when an audio segment derived from a particular user speaking the term is detected by the hotword detector. Additionally or alternatively, a specific term classified as a negative hotword may be device-specific such that hotword detection is suppressed when the audio segment derived from users speaking the term is detected by the hotword detector implemented on a particular device but not on other speech-enabled devices associated with the same user(s). This follows the notion that devices located in some environments may be more prone to hotword detection false positives than devices located in other environments due to acoustic variation, variation in vocabulary, and variation across users who typically speak in the environments.

(15) Referring to FIG. 1, in some implementations, an example system **100** includes a user device **102** associated with one or more users **10** and in communication with a remote system **110** via a network **104**. The user device **102** may correspond to a computing device, such as a mobile phone, computer (laptop or desktop), tablet, smart speaker/display, smart appliance, smart headphones, wearable, vehicle infotainment system, etc., and is equipped with data processing hardware **103** and memory hardware **105**. The user device **102** includes or is in communication with one or more microphones **106** for capturing utterances from the respective user **10**. The remote system **110** may be a single computer, multiple computers, or a distributed system (e.g., a cloud environment) having scalable/elastic computing resources **112** (e.g., data processing hardware) and/or storage resources **114** (e.g., memory hardware).

(16) The user device **102** includes a first stage hotword detector **210** (also referred to as a hotword

detection model) configured to detect the presence of a hotword in streaming audio **118** without performing semantic analysis or speech recognition processing on the streaming audio **118**. In some implementations, user device **102** also includes an initial coarse hotword detector **205** (FIG. 2) that is trained to initially detect the presence of the hotword before invoking the first stage (fine) hotword detector **210** to receive the streaming audio **118** and confirm whether or not the hotword is detected in the streaming audio **118**. In some implementations, the first stage hotword detector **210** includes a trained neural network (e.g., a memorialized neural network) received from the remote system **110** via the network **104**. The remote system **110** may push updates or new version of the first stage hotword detector **210** to a population of user devices **102** of a given local and/or associated with users of a specific language. That is, all user devices **102** associated with United States-English speaking users receive the same first stage hotword detector **210**, which may include a neural network trained to detect the presence of the hotword in streaming audio **118** when spoken by English speaking users in the United States.

(17) In some examples, the first stage hotword detector **210** executing on the user device **102** is configured to detect a presence of the hotword “Hey Google” in the streaming audio **118** to initiate a wake-up process on the user device **102** for processing the hotword and/or one or more other terms (e.g., query or command) following the hotword in the streaming audio **118**. The first stage hotword detector **210** may be configured to generate a probability score that indicates a presence of the hotword in audio features of the streaming audio **118** captured by the user device **102**, and detect the hotword in the streaming audio **118** when the probability score satisfies a hotword detection threshold of the first stage hotword detector **210**. Accordingly, the first stage hotword detector **210** may detect a hotword event in streaming audio **118** captured by the user device **102** when a probability score generated for audio features of the streaming audio **118** satisfy the hotword detection threshold.

(18) In the example shown, the user **10** speaks an utterance **119** that includes a term/phrase (e.g., “Hey Poodle”) captured as streaming audio **118** by the user device **102** that has a similar pronunciation as a fixed term/phrase (e.g., “Hey Google”) designated as the hotword the first stage hotword detector **210** is trained to detect. Notably, the user **10** may pronounce the term “Hey Poodle” such that it is more difficult for the first stage hotword detector to distinguish from the designated hotword “Hey Google” than it would be when spoken by other users that use a slightly different pronunciation. As a result, the first stage hotword detector **210** executing on the user device **102** falsely detects the presence of the hotword by generating a probability score for audio features associated with “Hey Poodle” that satisfies the hotword detection threshold, thereby triggering initiation of a wake-up process that was not intended by the user **10**.

(19) When the probability score for the audio features associated with “Hey Poodle” satisfy the hotword detection threshold, the first stage hotword detector **210** further transmits audio data **120** characterizing the hotword event to a second stage hotword detector **220** executing on the remote system **110**. In some examples, the audio data **120** is a direct representation of the streaming audio **118**, while in other examples, the audio data **136** represents the streaming audio **118** after processing by the first stage hotword detector **210** (e.g., to identify and/or isolate specific audio characteristics of the streaming audio **118** or to convert the streaming audio **118** to a format suitable for transmission and/or processing by the second stage hotword detector **220**). For instance, the audio data **120** includes a first segment **121** chopped from the streaming audio **120** that includes the relevant audio features associated with the presence of the hotword detected by the first stage hotword detector **210**. The audio data **120** also includes a second segment **122** that includes audio features captured by the user device **102** in the streaming audio **118** that follow the first segment **121**. Typically, the first segment **121** is generally of a fixed duration sufficient for containing audio features associated with the designated hotword. The second segment **122**, however, may have a variable duration containing audio captured by the user device **102** while the microphone **106** is open. The second segment **122** may capture a query-type utterance that requires further processing

(e.g., automated speech recognition and/or semantic interpretation) on one or more terms to identify a query or command in the audio data. In the example scenario, since the first segment **121** includes audio features captured by the user device that are not associated with the user speaking the hotword, but are rather associated with another term/phrase that the user **10** pronounces similar to the hotword, the user **10** did not intend to invoke the device **102** through speech, and therefore, the second segment **122** is likely not to include a query-type utterance, but instead include background noise captured in the streaming audio **118** from the environment of the user device **102**.

(20) The first stage hotword detector **210** and the second stage hotword detector **220** cooperate to form a cascade hotword detection architecture **200** whereby the second stage hotword detector **220** is configured to confirm whether or not a hotword detected by the first stage hotword detector **210** is present in the audio data **120**. Specifically, the second stage hotword detector **220** executing on the remote system **110** processes the audio data **120** to determine whether a hotword is detected by the second stage hotword detector in the first segment **121** of the audio data **120**. In some examples, the second stage hotword detector **210** is implemented as an automated speech recognition (ASR) engine that performs speech recognition on the first segment **121** of the audio data **120** to determine whether the hotword is present. The second stage hotword detector **210** may detect the presence of the hotword in the first segment **121** when a probability of recognizing the hotword satisfies a hotword detection threshold.

(21) In other examples, the second stage hotword detector **220** is similar to the first stage hotword detector **210** in that the second stage hotword detector **220** is a model implemented as a trained neural network configured to detect the presence of the hotword in the first audio segment **121** without performing semantic analysis or speech recognition processing. In these examples, the second stage hotword detector **220** may be associated with a larger version of the hotword detection model used by the first stage hotword detector **210** and include a different neural network that is potentially more computationally-intensive than the neural network of the first stage hotword detector **210**, thereby offering an increased hotword detection accuracy over the first stage hotword detector **210**, which is limited by resources of the user device **102**. The second stage hotword detector **220** may generate a probability score indicating the presence of the hotword in the first segment of audio data **120** and detect the presence of the hotword when the probability score satisfies a hotword detection threshold of the second stage hotword detector **220**. Here, a value of the hotword detection threshold at the second stage hotword detector **220** may be the same or different than a value of the hotword detection threshold at the first stage hotword detector **210**. In some examples, the value of the hotword detection threshold at the second stage hotword detector **220** is set higher to require the second stage hotword detector **220** to be more confident when determining whether or not a hotword is present in the first audio segment **121**.

(22) In some implementations, the second stage hotword detector **220** executes on the user device **102** (e.g., the data processing hardware **103**) to implement the entire cascade hotword detection architecture **200** on-device without the use of the remote system **110**. The second stage hotword detector **220** when executing on the user device **102** can be implemented as an on-device ASR engine to detect the presence of the hotword by performing speech recognition on the first audio segment **121** or as a larger version of the hotword detection model implemented by the first stage hotword detector **210** to detect the presence of the hotword in the first audio segment **121** without performing speech recognition.

(23) FIG. 2 provides an example of the cascade hotword detection architecture **200** of FIG. 1 including the first stage hotword detector **210**, the second stage hotword detector **220**, and optionally the initial coarse hotword detector **205**. In some examples, when the user device **102** is a battery-powered device, the data processing hardware **103** of the user device **102** collectively includes a first processor **60** (e.g., digital signal processor (DSP)) and a second processor **70** (e.g., application processor (AP)). The first processor **60** consumes less power while operating than the

second processor **70** consumes while operating. As used herein, the first processor **60** may be interchangeably referred to as a DSP and the second processor **70** may be interchangeably referred to as an “AP” or a “device SoC”. The initial coarse hotword detector **205** may run on the first processor **60** and the first stage hotword detector **210** may run on the second processor **70**. The second stage hotword detector **220** may execute on a server (e.g., remote system **110**) in communication with the user device **102** to provide server-side hotword confirmation that leverages increased processing capabilities at the server. Alternatively, the second stage hotword detector **220** may run on the second processor **70** of the user device **102** to implement the entire cascade hotword detection architecture on-device.

(24) Generally, the coarse hotword detector **205** resides on a specialized DSP **60**, includes a smaller model size than a model associated with the first stage hotword detector **210**, and is computationally efficient for coarsely screening input streaming audio **118** for hotword detection. Accordingly, the specialized DSP **60** e.g., first processor) may be “always-on” so that the coarse hotword detector **205** is always running to coarsely screen for hotword candidates in multi-channel audio **118**, while all other components of the user device **102**, including the main AP **70** (e.g., second processor), are in a sleep state/mode to conserve battery life. On the other hand, the first stage hotword detector **210** resides on the main AP **70**, includes a larger model size and provides more computational output than the coarse stage hotword detector **205** for providing a more accurate detection of the hotword that was initially detected by the coarse hotword detector **205**. Thus, the first stage hotword detector **210** may be more stringent in determining whether or not the hotword is present in the audio **118**. While the DSP **60** is “always-on”, the more power consuming main AP **70** operates in a sleep mode to reserve battery-life until the coarse hotword detector **205** at the DSP **60** detects the candidate hotword in the streaming audio **118**. Thus, only once the candidate hotword is detected, does the DSP **60** trigger the main AP **70** to transition from the sleep mode and into a hotword detection mode for running the first stage hotword detector **210**.

(25) Upon receiving the streaming audio **118**, the always-on DSP **60** executes/runs the coarse hotword detector **205** for determining whether a hotword is detected in the respective audio features of the streaming audio **118**. Notably, the AP **70** may operate in the sleep mode when the multi-channel audio is received at the DSP **60** and while the DSP **60** processes the respective audio features of the streaming audio **118**.

(26) When the coarse hotword detector **205** detects the hotword in the streaming audio **118**, the DSP **60** provides chopped audio data **120** to the AP **70**. In some examples, the DSP **60** providing the chopped audio data **120** to the AP **70** triggers/invokes the AP **70** to transition from the sleep mode to the hotword detection mode. In some implementations, the audio data **120** chopped from the streaming audio **118** includes a first segment **121** characterizing the hotword detected by the coarse hotword detector **205** in the streaming audio **118**. That is, the first includes a duration sufficient to safely contain the detected hotword. Additionally, audio data **212** includes a second segment **122** following the first segment **121** that may include a duration of audio containing a spoken query. The coarse hotword detector **205** is optional and the first stage hotword detector **210** may initially detect the hotword event in the streaming audio **118**, chop the audio data **120** including the first and second segments **121**, **122**, and provide the chopped audio data **120** to the second stage hotword detector **220**.

(27) When the hotword is detected by the first stage hotword detector **210** in the first segment **121** of the audio data **120**, the AP **70** initiates a wake-up process on the user device **102** and provides the audio data **120** to the second stage hotword detector **220** for processing to determine/confirm whether the hotword is detected by the second stage hotword detector **220** in the first segment **121** of the audio data **120**. In examples where the cascade hotword detection architecture **200** is implemented entirely on-device, the AP simply passes the audio data **120** to the second stage hotword detector **220** also running on the AP **70**. In other examples, where the second stage hotword detector **220** is implemented at a server **110**, the AP **70** instructs the user device **102** to

transmit the audio data **120** via a network **104** to the second stage hotword detector **220**.

(28) Referring back to FIG. **1**, in some examples, when the hotword is not detected by the second stage hotword detector **220** in the first segment **121** of the audio data **120** (i.e., the probability score satisfies the hotword detection threshold), a negative hotword classifier **300** executing on the remote system **110** classifies the first segment **121** of the audio data **120** as containing a negative hotword (e.g., Hey Poodle) that caused a false detection of the hotword event in the streaming audio **118** by the first stage hotword detector **210**. In configurations when the second stage hotword detector **220** runs on the user device **102**, the negative hotword classifier **300** may also execute on the user device **102**. The negative hotword classifier **300** may provide a classification result **170** to a negative hotword updater **400** that indicates the classification of the first segment **121** of the audio data **120** as containing the negative hotword. The classification result **170** may provide the probability score **171** generated by the second stage hotword detector **220** for the first audio segment **121** and/or other pertinent information **172** associated with the corresponding classification result **170** that may be useful for the negative hotword updater **400** for updating the first stage hotword detector **210**. The other pertinent information **172** may include, without limitation, a transcript of the utterance **119**, a speaker identification score (e.g., a speaker embedding) identifying the speaker characteristics of the user **10** that spoke the utterance **119**, a time stamp of the hotword event indicating the time of day and/or a day of the week, and a negative hotword confidence score **304** (FIG. **3**) indicating a confidence for classifying the first segment **121** as the negative hotword. In the example shown, the negative hotword updater **400** executes on the user device **102** for personalizing hotword detection on the user device **102** by updating the first stage hotword detector **210** to prevent triggering the hotword event in subsequent audio data that contains the negative hotword.

(29) Sending the classification result **170** to the negative hotword updater **400** may cause the negative hotword updater **400** to update the first stage hotword detector **210** to prevent triggering the hotword event in subsequent audio data that contains the negative hotword (e.g. Hey Poodle). In some implementations, when the hotword is not detected by the second stage hotword detector **220** in the first segment **121** of the audio data **120**, the remote system **110** (or the user device **102**) suppresses a wake-up process on the user device **102** for processing the hotword and/or one or more other terms following the hotword in the streaming audio **118**. In some implementations, the remote system **110** suppresses the wake-up process by sending suppression instructions **160** to the user device **102** that causes the user device **102** to suppress the wake-up process. In other implementations, the providing of the classification result **170** indicating that the first segment **121** of the audio data **120** contains the negative hotword causes the user device **102** to suppress the wake-up process. In yet other implementations, the remote system **110** suppresses the wake-up process by not responding to the user device **102** (e.g., by closing the network connection) after receiving the audio data **120**. A lack of response from the remote system **110** may cause the user device **102** to suppress the wake-up process. That is, the user device **102**, in some examples, only initiates the wake-up process upon receiving confirmation from the second stage hotword detector **220** that the hotword was present in the streaming audio **118**. The user device **102** may independently suppress the wake-up process. For example, when the query or command following the hotword is empty, the user device **102** may automatically suppress the wake-up process (i.e., the streaming audio **118** following the hotword fails to include a command or query directed at the user device **102**).

(30) In some scenarios, after suppressing the wake-up process due to the second stage hotword detector **220** not detecting the presence of the hotword in the first segment **121** of the audio data **120**, the negative hotword classifier **300** determines whether an immediate follow-up query was provided by the user **10** of the user device **102**. This determination may be made when no subsequent hotword event detected by the first stage hotword detector **210** is received by the second stage hotword detector **220**. Here, a determination that the user **10** did not provide an

immediate follow-up query serves as additional confirmation that the user **10** did not previously intend to speak the hotword in the utterance **119**, but rather spoke a term (“Hey Poodle”) with a similar pronunciation to the particular term/phrase (“Hey Google”) designated as the hotword. Accordingly, classifying the first segment **121** of the audio data **120** as containing the negative hotword may be further based on a determination that no follow-up query was received from the user device **102** after suppressing the wake-up process.

(31) In additional examples, when the second stage hotword detector **220** detects the presence of the hotword (“Hey Google”) in first segment **121** of the audio data **120** (i.e., the probability score satisfies the hotword detection threshold) despite the user **10** really speaking another similarly sounding phrase (“Hey Poodle”), the second segment **122** (and optionally the first segment **121**) of the audio data **120** are provided to the query processor **180**. Here, the query processor **180** processes the second segment **122** of the audio data **120** to determine whether the second segment **122** of the audio data **120** is indicative of a spoken query-type utterance. In examples when the second stage hotword detector **220** is implemented as an ASR engine, the query processor **180** process the resulting speech recognition result by performing semantic analysis to determine if the second segment **122** is indicative of the query-type utterance. In other examples, when the second stage hotword detector **220** is implemented as a hotword detection model, the query processor **180** is implemented as the ASR engine that processes the second segment **122** of the audio data **120** by performing speech recognition and then performing semantic analysis on the speech recognition result. As used herein, a query-type utterance corresponds to an utterance that was directed to the user device **102**, e.g., an utterance directed to a digital assistant interface for querying the digital assistant to perform an operation or action. Thus, when the second segment **122** of the audio data **120** is indicative of the query-type utterance, there is a strong likelihood that the second stage hotword detector **220** was accurate in detecting the presence of the hotword in the first segment **121** of the audio data **120**. Otherwise, when the query processor **180** determines that the second segment **122** is not indicative of the query-type utterance, there exists a strong likelihood that the second stage hotword detector **220** was incorrect in detecting the presence of the hotword in the first segment **121**.

(32) The query processor **180** may provide a score **182** indicating whether or not the second segment **122** is indicative of a query-type utterance. In some examples, the score **182** is binary where the score **182** of zero or one (1) indicates the query-type utterance and the score of the other one of zero or one (1) does not indicate the query-type utterance. In other examples, the score **182** provides a likelihood (e.g., probability) that the second segment **122** is indicative of a query-type utterance. Here, when the score **182** fails to satisfy a query-type utterance threshold the second segment **122** may not be indicative of the query-type utterance. In the example shown, the negative hotword classifier **300** may receive the score **182** as an input in addition to the determination made by the second stage hotword detector **220** for determining whether or not the first segment **121** of the audio data **120** should be classified as containing the negative hotword.

(33) Thus, when the negative hotword classifier **300** receives an indication from the query processor **180** that the second audio segment **122** of the audio data is not indicative of the spoken query-type utterance, the negative hotword classifier **300** may classify the first segment **121** of the audio data **120** as containing the negative hotword to indicate that the second stage hotword detection **220** provided a false acceptance. The negative hotword classifier **300** may additionally receive the probability score **171** generated by the second stage hotword detector **220** for the first segment **121**, whereby a probability score only satisfying the hotword detection threshold by a narrow margin may further bias the negative hotword classifier **300** to classify the first segment **121** as containing the negative hotword. Moreover, after the query processor **180** determines that the second segment **122** is not indicative of the query-type utterance, the negative hotword classifier **300** may also determine whether an immediate follow-up query was provided by the user **10** of the user device. As discussed above, a determination that the user **10** did not provide an

immediate follow-up query serves as additional confirmation that the user **10** did not previously intend to speak the hotword in the utterance **119**, but rather spoke a term (“Hey Poodle”) with a similar pronunciation to the particular term/phrase (“Hey Google”) designated as the hotword. Accordingly, classifying the first segment **121** of the audio data **120** as containing the negative hotword may be further based on the determination that no follow-up query was received from the user device **102**.

(34) In some examples, after receiving the audio data **120** characterizing the hotword event detected by the first stage hotword detector **210**, the remote system **110** receives a negative user interaction **162** indicating user suppression of a wake-up process on the user device **102**. That is, the false acceptance instance by the first stage hotword detector **210** in detecting the hotword event when the user spoke the negative hotword “Hey Poodle” may trigger the user device **102** to initially wake-up while waiting for the second stage hotword detector **220** to confirm or reject the presence of the hotword. Here, the user device **102** may provide an audible and/or visual notification to inform the user that the user device **102** is awake and the user **10** may provide a negative user interaction **162** to revert the device **102** back to the sleep state since the user **10** did not intend to trigger the wake-up process. For instance, the user **10** may press a physical button on the user device, provide a gesture, or, when the user device **102** includes a display, select a graphic rendered in a graphical user interface displayed on the display that causes the user device **102** to revert back to the sleep state. In some implementations, the negative hotword classifier **300** uses the negative user interaction **162** indicating user suppression of the wake-up process on the user device as an input for classifying the first segment **121** of the audio data **120** as containing the negative hotword.

(35) In some additional examples, when the query processor **180** determines that the second segment **122** of the audio data **120** is indicative of the spoken query-type utterance, the query processor **180** provides a query **185** to a search engine **190** (or other downstream application) that contains a transcription of the second segment **122** of the audio data **120**. Here, the search engine **190** provides results **192** responsive to the query **185** back to the user device **102**. Here, the query processor **180** may have identified the second segment **122** as being indicative of a query-type utterance even though the second segment **122** of the audio data **120** corresponded to background speech or other background audio that was captured by the user device **102** in streaming audio **118** after the first stage hotword detector **210** detected the false positive hotword event when the user spoke “Hey Poodle”. This background audio may be captured in the streaming audio **118** and the query processor **180** may identify a query-type utterance and provide a corresponding query **185** to the search engine **190** to obtain a result **192**. The result **192** may be audibly and/or visually output by the user device **102** to the user **10** even though the user **10** never intended to invoke the user device **102**. As a result, the user **10** may provide the negative user interaction **162** indicating that the user **10** negatively interacted with the results **192**. For instance, the user **10** may provide a spoken input indicating the user **10** is confused with the results or a statement that the user **10** did not provide a query. Additionally or alternatively, the user **10** may provide an input indication indicating an instruction/command to dismiss the results **192**.

(36) In other scenarios, the result **192** is a prompt from the digital assistant stating for audible output from the device **102** that the user **10** needs to provide confirmation for performing an action, e.g., “You asked for the current weather, is that correct?”, whereby the negative user interaction can be the user **10** speaking “No, I did not ask about the weather”. Similarly, the result **192** can be a prompt requesting the user to repeat a query because the query processor **180** was not confident in the query, e.g., “I did not understand your question, please repeat?”, whereby the negative user interaction can be the user **10** expressing confusion by uttering “Huh”, the user **10** affirmatively dismissing the prompt by speaking “I did not ask anything”, or simply the user not responding within a predetermined period of time. Thus, the negative user interaction **162** may be provided to the negative hotword classifier **300** in addition to one or more of the other inputs discussed above such as an indication that the hotword was not detected by the second stage hotword detector **220**.

in the first segment of the audio data **120**, an indication that the second segment **122** of the audio data **120** is not associated with a query-type utterance, or an indication that the user **10** did provide an immediate follow-up query after the wake-up process was suppressed.

(37) FIG. **3** shows an example of the negative hotword classifier **300** of FIG. **1** receiving one or more input features **302** for making the determination of whether or not a first segment **121** of audio data **120** should be classified as a negative hotword. When the negative hotword classifier **300** determines that the first segment **121** of the audio should be classified as the negative hotword based on the one or more input features **302**, the negative hotword classifier **300** will generate the classification results **170** indicating classification of the first segment **121** of audio data **120** as the negative hotword as discussed above in FIG. **1**. The one or more input features **302** received by the negative hotword classifier **300** may include, without limitation, whether or not the second hotword detector **220** detected the presence of the hotword in the first segment **121** of audio data **120** and/or the probability score **171**, whether an immediate follow-up query was received from the user device **102**, whether or not the second segment **122** of the audio data **120** is provided an indication a score **182** indicating whether or not the second segment **122** includes a query-type utterance (e.g., by providing the score **182** indicating whether or not the second segment **122** is indicative of a query-type utterance), a transcription of the first segment **121** and/or the second segment **122** of audio data **120**, and whether or not a negative user interaction **162** is received indicating user suppression of a wake-up process on the user device **102** and/or that the user **10** negatively interacted with results **192** responsive to processing the second segment **122** (and/or optionally the first segment **121**) of the audio data **120** as a query-type utterance (e.g., providing the query **185** to the search engine **190** or other downstream application).

(38) Some input features **302** may be weighted more heavily when determining whether the first segment **121** of the audio data **120** should be classified as the negative hotword. For instance, the second stage hotword detector **220** failing to detect the presence of the hotword in the first segment **121** is a strong indication that the first segment **121** includes a negative hotword that caused the false acceptance instance at the first stage hotword detector **210**. The magnitude of the probability score **171** may bias the classification result **170**. For instance, the probability score **171** failing to satisfy the hotword detection threshold at the second stage hotword detector **220** by a wide margin provides a greater likelihood of a negative hotword than if the probability score **171** missed satisfying the hotword detection threshold by small margin.

(39) In some configurations, the negative hotword classifier **300** includes a trained classifier (which may include a neural network model trained via machine learning) configured to generate a negative hotword confidence score **304** indicating a likelihood that the first segment **121** of the audio data **120** includes a negative hotword. The classifier **300** may classify the first segment **121** as containing the negative hotword when the negative hotword confidence score **304** satisfies a confidence threshold. The negative hotword confidence score **304** may be included in the classification results **170** received by the negative hotword updater **400** of FIG. **1** for use in updating the first stage hotword detector **210** to not detect the hotword event in subsequent audio containing the negative hotword. In some examples, the negative hotword confidence score **304** is a binary score indicating that the first segment **121** of audio data **120** includes the negative hotword, and thus should be classified as the negative hotword, or that the first segment **121** does not include the negative hotword.

(40) Referring to FIGS. **1** and **4**, in some examples, the negative hotword updater **400** updates the first stage hotword detector **210** to prevent triggering the hotword event in subsequent audio data that includes a negative hotword by providing the classification result **170** that includes the first segment **121** of the audio data **120** to the user device **102**. Here, the user device **102** may be configured to retain the first stage hotword detector **210** using the first segment **121** of the audio data **120** classified as containing the negative hotword. For example, the first segment **121** of audio data **120** may be labeled as a negative hotword and provided as a training input to the first stage

hotword detector **210** so the first stage hotword detector **210** learns to not detect the presence of the hotword (“Hey Google”) in subsequent audio data containing the negative hotword (“Hey Poodle”). As used herein, retraining the first stage hotword detector **210** may include retraining an existing hotword detector **210** running on the user device **102** that initially detected the hotword event incorrectly or may include a new first stage hotword detector **210** pushed to the user device **102** at a later time.

(41) Moreover, updating the first stage hotword detector **210** may also include updating the optional initial coarse hotword detector **205** running on a DSP **60** (FIG. 2) when the user device **102** includes a battery-powered device. As with the first stage hotword detector **210**, updating the initial coarse hotword detector **205** may include retraining the initial coarse hotword detector **205** to prevent triggering hotword events for audio data including a negative hotword. In some examples, only the initial coarse hotword detector **205** is updated to prevent triggering the hotword event in subsequent audio data that includes a negative hotword.

(42) As shown in FIG. 4, the negative hotword updater **400** executes on the user device **102** store each instance of a first segment **121** of audio data **120** classified by the negative hotword classifier **300** as containing a corresponding negative hotword in the memory hardware **105**. In the example shown, the user device **102** stores each instance where a first segment **121** of audio data **120** was classified as a negative hotword by storing the corresponding classification result **170** for each instance. Here, the classification result **170** includes the first segment **121** classified as containing the negative hotword, the probability score **171** indicating the likelihood that the first segment **121** included the actual hotword, and the other pertinent information **172**, such as the transcript of the utterance **119**, a speaker identification score (e.g., a speaker embedding) identifying the speaker characteristics of the user **10** that spoke the utterance **119**, a time stamp of the hotword event indicating the time of day and/or a day of the week, and a negative hotword confidence score **304** (FIG. 3) indicating a confidence for classifying the first segment **121** as the negative hotword. In the example shown, the user device **102** stores one or more classification results **170** each associated with a different corresponding negative hotword. For instance, one more classification results **170Aa-n**, **170Ba-n**, **170Ca-n** may be stored for each of the negative hotwords “Poodle”, “Doodle”, and “Noodle” that when spoken by the user **10**, are pronounced similar to the designated hotword “Hey Google”.

(43) In some implementations, the user device **102** (via the negative hotword updater **400**) is configured to retrain the first stage hotword detector **210** based on an aggregation of the number of instances (e.g., number of classification results **170**) of the first segment **121** of the audio data **120** classified as containing the negative hotword stored in the memory hardware **105**. Here, the number of instances of audio data classified as containing the same hotword satisfying a threshold number of instances may establish a pattern that the user **10** is regularly speaking the negative hotword falsely detected as the designated hotword by the first stage hotword detector **210**. In some examples, the user device **102** requires a specified number of false acceptance instances resulting from the user **10** speaking the same term when the negative hotword confidence scores **304** associated with scores is relatively low, e.g., the negative hotword confidence scores **304** only satisfied the threshold by a narrow margin.

(44) With continued reference to FIG. 4, in some examples, the negative hotword updater **400** may append an embedding representation **12** to each classification result **170** stored in the memory hardware **105**. Here, the first stage hotword detector **210** may compute an embedding representation **12** for any audio data characterizing a hotword event detected by the first stage hotword detector **210**, and when the audio data **120** is subsequently classified by the negative hotword classifier **300** as containing a negative hotword, the negative hotword updater **400** may append the embedding representation **12** to the corresponding instance of the classification result **170**. In some implementations, the negative hotword updater **400** aggregates/averages the embedding representations **12** stored in the memory hardware **105** for each corresponding negative

hotword to generate a reference embedding **15** for each corresponding negative hotword. For example, a corresponding reference embedding **15** may be generated for each of the negative hotword “Poodle”, “Noodle”, and “Doodle”.

(45) FIG. 5 shows a schematic view **500** depicting an example where user device **102** captures subsequent audio data **120** corresponding to another utterance **519** spoken by the user **10** that includes the term “My Poodle” and that causes the first stage hotword detector **210** to falsely detect another hotword event. The first stage hotword detector **210** executing on the user device **102** computes an evaluation embedding representation **18** for the subsequent audio data **120** (e.g., the portion of the subsequent audio data **120** characterizing the hotword event). The user device **102** may simultaneously access the classification result(s) **170** stored in the memory hardware **105** that may each include an embedding representation **12** of the corresponding first segment **121** of audio data **120** that was classified as one of the negative hotwords (e.g., “Poodle”, “Noodle”, and “Doodle”). Additionally or alternatively, the user device **102** may access a corresponding reference embedding **15** generated for each of the negative hotwords as described above with reference to FIG. 4.

(46) In some implementations, a scorer **510** compares the evaluation embedding representation **18** computed for the subsequent audio data **120** with all of the reference embeddings **15** generated and stored for each of the negative hotwords. In these implementations, the reference embedding representations **12**, **15** associated with each negative hotword “Poodle”, “Doodle”, and “Noodle” will all be clustered together in an embedding representation space distinct from the clusters of the reference embeddings associated with the other negative hotwords. Accordingly, the scorer **510** may determine a similarity score **515** between the reference embedding representation **12** computed for each first segment **121** of audio data **120** that was classified as any of the negative hotword and the evaluation embedding representation **18** for the subsequent audio data **120**. Additionally or alternatively, the scorer **510** may determine a similarity score **515** between each corresponding reference embedding representation **15** that represents an aggregate/average embedding representation for a corresponding one of the negative hotwords (e.g., “Poodle”, “Doodle”, and “Noodle”). In some examples, each similarity score **515** is associated with a distance (e.g., a cosine distance) between the evaluation embedding representation **18** and the reference embedding representation **12**, **15** in the embedding representation space.

(47) After the scorer **510** determines/generates the similarity score(s) **515**, a classifier **520** may compare each similarity score **515** to a similarity score threshold and determine/classify the subsequent audio data **120** as including the negative hotword when the similarity score **515** satisfies the similarity score threshold. In some examples, the similarity score threshold represents a maximum allowable cosine distance between embedding representations associated with a same negative hotword. In some scenarios, when a similarity score **515** is computed between the evaluation embedding **18** for the subsequent audio and a corresponding reference embedding representation **12** computed for each instance of a first segment **121** of audio data **120** being classified as a negative hotword, multiple similarity scores **515** may satisfy the similarity score threshold. To illustrate, in the example shown, the similarity scores **515** between the evaluation embedding **18** and the corresponding reference embedding representations **12** classified as the negative hotword “Poodle” will satisfy the similarity score threshold to indicate that the evaluation embedding **18** falls into a cluster of the embedding representations **12** classified as the negative hotword Poodle and outside the clusters of embedding representations **12** classified as the other negative hotwords “Doodle” and “Noodle”. Thus, when the classifier **520** determines that the similarity score **515** satisfies the similarity threshold, the classifier **520** determines that the subsequent audio data **120** includes the negative hotword. As a result, the classifier **520** may instruct the first stage hotword detector **210** to suppress detecting the hotword event in the subsequent audio data or instructs the user device **102** to revert back to a sleep state if the first stage hotword detector **210** affirmatively falsely detected the hotword event and triggered initiation of a

wake-up process on the user device.

(48) FIG. 6 provides a flowchart of example operations for a method **600** of personalizing a hotword detector on a user device based on classifying specific terms as negative hotwords which caused previous instances of hotword detection false positive by the hotword detector. At operation **602**, the method **600** includes receiving, at data processing hardware **103**, **112**, audio data **120** characterizing a hotword event detected by a first stage hotword detector **210** in streaming audio **118** captured by a user device **102**. The first stage hotword detector **210** may execute on a digital signal processor (DSP) of the data processing hardware **103** of the user device **102** or execute on an application processor of the data processing hardware **103** of the user device **102**.

(49) At operation **604**, the method **600** includes processing, by the data processing hardware **103**, **112**, using a second stage hotword detector **220**, the audio data **120** to determine whether a hotword is detected by the second stage hotword detector **220** in a first segment **121** of the audio data **120**. The second stage hotword detector **220** may be implemented as an ASR engine that performs automate speech recognition to determine whether the hotword is recognized in the first segment **121**. The second stage hotword detector **220** may be implemented as a hotword detection model in other configurations, whereby the hotword detection model determines whether or not the hotword is detected in the first segment **121** without performing speech recognition.

(50) At operation **606**, when the hotword is not detected by the second stage hotword detector **220** in the first segment **121** of the audio data **120**, the method **600** includes classifying, by the data processing hardware **103**, **112**, the first segment **121** of the audio data **120** as containing a negative hotword that caused a false detection of the hotword event in the streaming audio **118** by the first stage hotword detector **210**. At operation **608**, the method **600** includes updating, by the data processing hardware **103**, **112**, the first stage hotword detector **210** to prevent triggering the hotword event in subsequent audio data **120** that contains the negative hotword.

(51) A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an “application,” an “app,” or a “program.” Example applications include, but are not limited to, system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

(52) FIG. 7 is schematic view of an example computing device **700** that may be used to implement the systems and methods described in this document. The computing device **700** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document.

(53) The computing device **700** includes a processor **710**, memory **720**, a storage device **730**, a high-speed interface/controller **740** connecting to the memory **720** and high-speed expansion ports **750**, and a low speed interface/controller **760** connecting to a low speed bus **770** and a storage device **730**. Each of the components **710**, **720**, **730**, **740**, **750**, and **760**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **710** can process instructions for execution within the computing device **700**, including instructions stored in the memory **720** or on the storage device **730** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **780** coupled to high speed interface **740**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **700** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(54) The memory **720** stores information non-transitorily within the computing device **700**. The memory **720** may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **720** may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by the computing device **700**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

(55) The storage device **730** is capable of providing mass storage for the computing device **700**. In some implementations, the storage device **730** is a computer-readable medium. In various different implementations, the storage device **730** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **720**, the storage device **730**, or memory on processor **710**.

(56) The high speed controller **740** manages bandwidth-intensive operations for the computing device **700**, while the low speed controller **760** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **740** is coupled to the memory **720**, the display **780** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **750**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **760** is coupled to the storage device **730** and a low-speed expansion port **790**. The low-speed expansion port **790**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

(57) The computing device **700** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **700a** or multiple times in a group of such servers **700a**, as a laptop computer **700b**, or as part of a rack server system **700c**.

(58) Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(59) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory, Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or

data to a programmable processor.

(60) The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(61) To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

(62) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A computer-implemented method executed on data processing hardware that causes the data processing hardware to perform operations comprising: receiving audio data characterizing a hotword detected by a first stage hotword detector in a first segment of the audio data; detecting, using a second stage hotword detector, the hotword in the first segment of the audio data, the second stage hotword detector different from the first stage hotword detector; based on detecting the hotword using the second stage hotword detector, processing a second segment of the audio data that follows the first segment of the audio data to determine the second segment of the audio data is not indicative of a spoken query-type utterance, the second segment of the audio data characterizing one or more other terms spoken after the hotword; and based on determining the second segment of the audio data is not indicative of the spoken query-type utterance: determining a negative hotword confidence score classifying the first segment of the audio data as a negative hotword; and updating the first stage hotword detector to prevent triggering a hotword event in subsequent audio data that contains the negative hotword based on the negative hotword confidence score.

2. The computer-implemented method of claim 1, wherein the operations further comprise: after receiving the audio data characterizing the hotword detected by the first stage hotword detector, determining that no follow-up query was provided by a user of a user device, wherein determining the negative hotword confidence score is further based on determining that no follow-up query was provided by the user of the user device.
3. The computer-implemented method of claim 1, wherein the operations further comprise: after receiving the audio data characterizing the hotword detected by the first stage hotword detector, receiving a user interaction indicating user suppression of a wake-up process on a user device, wherein determining the negative hotword confidence score is further based on the user interaction indicating user suppression of the wake-up process.
4. The computer-implemented method of claim 1, wherein updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data comprises providing the first segment of the audio data classified as containing the negative hotword to a user device, the user device configured to retrain the first stage hotword detector using the first segment of audio data classified as containing the negative hotword.
5. The computer-implemented method of claim 4, wherein the user device is configured to retrain the first stage hotword detector by: storing, in memory hardware of the user device, each instance of the first segment of the audio data classified as containing the negative hotword in memory hardware of the user device; and retraining the first stage hotword detector based on an aggregation of a number of instances of the first segment of the audio data classified as containing the negative hotword stored in the memory hardware.
6. The computer-implemented method of claim 5, wherein the user device is further configured to, prior to retraining the first stage hotword detector: determine that a corresponding confidence score associated each instance of the first segment of the audio data classified as containing the negative hotword fails to satisfy a negative hotword threshold score; and determine that the number of instances exceeds a threshold number of instances.
7. The computer-implemented method of claim 1, wherein: updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data comprises providing the first segment of the audio data classified as containing the negative hotword to a user device, the user device configured to: obtain an embedding representation of the first segment of the audio data; and store, in memory hardware of the user device, the embedding representation of the first segment of the audio data; and the user device is configured to determine when subsequent audio data characterizing the hotword event detected by the first stage hotword detector includes the negative hotword by: computing an evaluation embedding representation for the subsequent audio data; determining a similarity score between the embedding representation of the first segment of the audio data classified as the negative hotword and the evaluation embedding representation for the subsequent audio data; and when the similarity score satisfies a similarity score threshold, determining that the subsequent audio data includes the negative hotword.
8. The computer-implemented method of claim 1, wherein: the data processing hardware resides on a server in communication with a user device.
9. The computer-implemented method of claim 1, wherein: the data processing hardware resides on a user device associated with the user; and the first stage hotword detector executes on the data processing hardware.
10. The computer-implemented method of claim 1, wherein: the first stage hotword detector that executes on a digital signal processor (DSP) of the data processing hardware; and the second stage hotword detector that executes on an application processor of the data processing hardware.
11. A system comprising: data processing hardware; and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising: receiving audio data characterizing a hotword detected by a first stage hotword detector in a first

segment of the audio data; detecting, using a second stage hotword detector, the hotword in the first segment of the audio data, the second stage hotword detector different from the first stage hotword detector; based on detecting the hotword using the second stage hotword detector, processing a second segment of the audio data that follows the first segment of the audio data to determine the second segment of the audio data is not indicative of a spoken query-type utterance, the second segment of the audio data characterizing one or more other terms spoken after the hotword; and based on determining the second audio segment of the audio data is not indicative of the spoken query-type utterance: determining a negative hotword confidence score classifying the first segment of the audio data as a negative hotword; and updating the first stage hotword detector to prevent triggering a hotword event in subsequent audio data that contains the negative hotword based on the negative hotword confidence score.

12. The system of claim 11, wherein the operations further comprise: after receiving the audio data characterizing the hotword detected by the first stage hotword detector, determining that no follow-up query was provided by a user of a user device, wherein determining the negative hotword confidence score is further based on determining that no follow-up query was provided by the user of the user device.

13. The system of claim 11, wherein the operations further comprise: after receiving the audio data characterizing the hotword detected by the first stage hotword detector, receiving a user interaction indicating user suppression of a wake-up process on a user device, wherein determining the negative hotword confidence score is further based on the user interaction indicating user suppression of the wake-up process.

14. The system of claim 11, wherein updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data comprises providing the first segment of the audio data classified as containing the negative hotword to a user device, the user device configured to retrain the first stage hotword detector using the first segment of audio data classified as containing the negative hotword.

15. The system of claim 14, wherein the user device is configured to retrain the first stage hotword detector by: storing, in memory hardware of the user device, each instance of the first segment of the audio data classified as containing the negative hotword in memory hardware of the user device; and retraining the first stage hotword detector based on an aggregation of a number of instances of the first segment of the audio data classified as containing the negative hotword stored in the memory hardware.

16. The system of claim 15, wherein the user device is further configured to, prior to retraining the first stage hotword detector: determine that a corresponding confidence score associated each instance of the first segment of the audio data classified as containing the negative hotword fails to satisfy a negative hotword threshold score; and determine that the number of instances exceeds a threshold number of instances.

17. The system of claim 11, wherein: updating the first stage hotword detector to prevent triggering the hotword event in subsequent audio data comprises providing the first segment of the audio data classified as containing the negative hotword to a user device, the user device configured to: obtain an embedding representation of the first segment of the audio data; and store, in memory hardware of the user device, the embedding representation of the first segment of the audio data; and the user device is configured to determine when subsequent audio data characterizing the hotword event detected by the first stage hotword detector includes the negative hotword by: computing an evaluation embedding representation for the subsequent audio data; determining a similarity score between the embedding representation of the first segment of the audio data classified as the negative hotword and the evaluation embedding representation for the subsequent audio data; and when the similarity score satisfies a similarity score threshold, determining that the subsequent audio data includes the negative hotword.

18. The system of claim 11, wherein: the data processing hardware resides on a server in

communication with a user device.

19. The system of claim 11, wherein: the data processing hardware resides on a user device associated with the user; and the first stage hotword detector executes on the data processing hardware.

20. The system of claim 11, wherein: the first stage hotword detector that executes on a digital signal processor (DSP) of the data processing hardware; and the second stage hotword detector that executes on an application processor of the data processing hardware.
